

Cell Phone Radiation Safety

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Abstract

This report investigates the safety of cell phone radiation and cell towers by analyzing the physical properties of radio frequency (RF) electromagnetic waves. We first establish that cell phone signals, operating between 0.8 GHz and 2.6 GHz, are non-ionizing and carry energies many orders of magnitude below the ionization threshold. We then examine whether cell towers pose a meaningful health risk, finding that the power absorbed from a cell tower is negligible compared to that of a phone held directly to the body. Finally, we investigate resonance effects: while RF radiation at 80–100 MHz can resonate with the whole human body, cell phone frequencies are too high for large-scale body resonance, and their specific absorption rates (SAR) fall well within the FCC safety limit of 1.6 W/kg. We conclude that while current evidence does not indicate a significant health risk from normal cell phone use, more long-term research on proximity to cell towers is warranted.

Cell Phone Signal Basics

Cell phone signals are one of the many forms of electromagnetic radiation, a special form of energy that travels through space as a wave made of interwoven electric and magnetic fields. According to the Health Physics Society, cell phone signals and other signals used for

telecommunication services in general are radio frequency (RF) electromagnetic waves. These occupy the portion of the electromagnetic spectrum between 3 kHz and 300 GHz. According to the fundamental wave equation,

$$\lambda = \frac{c}{f}, \quad c = 3.0 \times 10^8 \text{ m/s},$$

this frequency range corresponds to a wavelength range spanning from 100 km to 1 mm [1].

Although electromagnetic waves can ionize human tissues and potentially cause cancer, we are not worried about the effect of phone signals on human health. This is because the energy range of phone signals is millions of orders of magnitude smaller than the lowest frequencies that can ionize biological material. According to the U.S. Department of Health and Human Services, ionization starts occurring at frequencies above the ultraviolet (or at least above the high end of ultraviolet). The frequencies below are classified as non-ionizing [2].

A UV light photon carries between 3.10 eV and 12.4 eV, whereas the quanta of a cell phone signal's energy range between

$$E_{\min} = hf_{\min} = (4 \times 10^{-15} \text{ eV/Hz})(3000 \text{ Hz}) = 1.2 \times 10^{-11} \text{ eV}$$

and

$$E_{\max} = hf_{\max} = (4 \times 10^{-15} \text{ eV/Hz})(300 \text{ GHz}) = 1.2 \times 10^{-3} \text{ eV}.$$

So, cell phone signals' energies are many orders of magnitude too low below the ionization threshold.

Although phone signals are not a probable cause of cancer, they can still cause some health issues if the intensity is too high. This is because radio frequency electromagnetic waves tend to heat animal tissues. So, if exposed in large quantities, the tissue may not have enough time to dissipate the heat, which can result in tissue damage. The intensity of phone signals is, however, too low to induce significant tissue damage [1].

Are Cell Towers Dangerous?

You could reduce your radiation exposure by being close to a cell tower because your phone does not need to use as much power to get a strong signal near a cell tower. On the other hand, when you are farther away from a cell tower, your phone needs to increase the power it is releasing to keep up with the level of signal it would have if it were near a cell tower. Though cell towers also release radiation themselves, the radiation from your phone is more significant in your personal radiation exposure level.

When imagining the entire area of a cell phone collecting signal from a tower, there are different ranges of intensities. The signal strength range is -30 dBm to -110 dBm. A signal strength of -30 dBm is a strong signal, and -110 dBm is a weak signal; the closer the signal strength is to 0, the stronger the signal will be. Using the decibel-milliwatt to intensity conversion,

$$\beta_{\text{dBm}} = 10 \text{ dBm} \cdot \log_{10}\left(\frac{P}{P_0}\right), \quad P_0 = 10^{-3} \text{ W},$$

the intensities that correspond to those signal strengths are:

$$I_{-30} = 1 \times 10^{-15} \text{ W/m}^2, \quad I_{-110} = 1 \times 10^{-23} \text{ W/m}^2.$$

A cell phone 1 km away from a tower would need to transmit approximately 1.26×10^{-16} W of power to receive the minimum signal (-110 dBm), computed via the inverse-square law:

$$P = 4\pi d^2 \cdot I_{-110} = 4\pi(1000 \text{ m})^2 \cdot (1 \times 10^{-23} \text{ W/m}^2) = 1.2566 \times 10^{-16} \text{ W}.$$

With this power, a cell phone would need to be only 0.0999 m away from the cell tower to receive a signal at the strong end of the spectrum (-30 dBm), found by solving

$$d = \sqrt{\frac{P}{4\pi \cdot I_{-30}}} = \sqrt{\frac{1.2566 \times 10^{-16} \text{ W}}{4\pi \cdot (1 \times 10^{-15} \text{ W/m}^2)}} \approx 0.0999 \text{ m}.$$

For a phone to successfully receive a signal from a cell tower, the minimum intensity required is $1 \times 10^{-23} \text{ W/m}^2$. The minimum intensity required at the cell tower itself would be smaller

still, because cell phone towers have very large antennas that are sensitive to signals and are therefore able to pick up on small intensities. This is why cell towers are usually found spread out: they are able to emit and receive signals over large areas.

Despite their ability to cover large areas, cell towers do not emit a lot of power that is absorbed by humans. Even at the high end of the signal strength range (-30 dBm), the power released by a cell phone tower is only about $0.000\,001$ W:

$$P = 10^{-30 \text{ dBm}/10 \text{ dB}} \cdot 10^{-3} \text{ W} = 10^{-3} \cdot 10^{-3} \text{ W} = 1 \times 10^{-6} \text{ W}.$$

The maximum specific absorption rate for a phone in the United States is 1.6 W/kg of body weight [3]. Taking one-quarter of this gives 0.4 W — the estimated power output of a cell phone’s signal transmitter — which is 400,000 times the $0.000\,001 \text{ W}$ emitted by a cell tower. Absorbing the maximum power from a cell tower is therefore essentially negligible by comparison.

When reflecting on the Veritasium video’s assessment, we are confident that being closer to a cell tower with a cell phone is safer than being far from one, because your cell phone does not have to emit stronger signals to reach the tower. However, we do not fully know the long-term effects of being close to a cell tower. Since cell towers are emitting very strong signals, being close to them could be as negatively impacting on the body as being far away from one while your phone emits strong signals. On either side of the cycle, there are caveats to the safety of the distance between you and your device or cell tower.

Resonance and Radiofrequency Radiation

The Health Physics Society article “Radiofrequency (RF) Radiation” by Kelly Classic [1] mentions that a standing adult can absorb RF energy most efficiently when the frequency is between 80 and 100 MHz. This is described as a resonance condition, meaning that the wavelength of the radiation maximizes absorption. To understand this, we can model human tissue as mostly water and use its index of refraction, $n \approx 8.85$, for this frequency range.

Using the relationship $v = c/n$, the effective speed of light in the body is about:

$$v = \frac{3 \times 10^8 \text{ m/s}}{8.85} \approx 3.4 \times 10^7 \text{ m/s}.$$

The resonant wavelength in tissue is then:

$$\lambda_{80} = \frac{v}{f} = \frac{3.4 \times 10^7 \text{ m/s}}{80 \text{ MHz}} = 0.42 \text{ m}, \quad \lambda_{100} = \frac{v}{f} = \frac{3.4 \times 10^7 \text{ m/s}}{100 \text{ MHz}} = 0.34 \text{ m}.$$

So, the resonant wavelength lies between 34 and 42 cm, which corresponds to about one quarter of an average adult's height. This explains why whole-body resonance appears around these frequencies: the body in this case acts like an antenna, and when its size matches a fraction of the wavelength, the oscillating electric field can set up standing-wave patterns that enhance energy absorption.

However, cell-phone frequencies are roughly between 0.8 and 2.6 GHz, with correspondingly shorter wavelengths of roughly 12 to 37 cm. They do not match the dimensions of the whole human body. At these frequencies, the energy coupling between the electromagnetic field and the body is far less efficient, as the wave oscillates too quickly for large-scale resonance, so only a small fraction of the field's energy is absorbed by tissue. As a result, the specific absorption rate (SAR) — defined as the power absorbed per kilogram of body tissue — is very low:

$$\frac{0.001 \text{ W/kg (whole body from phone)}}{1 \text{ W/kg (whole body at resonance)}} = 10^{-3}.$$

This is well within the FCC's safety limit of 1.6 W/kg averaged over 1 g of tissue [4].

The same physics from the Veritasium video “*Microwaving Grapes Makes Plasma*” — where microwaves form resonant hotspots inside grapes when the wavelength fits the grape's size — applies to the human body: resonance depends on geometry and wavelength. Scaling that concept down, smaller body parts have their own possible resonances. Using the standing wave model for an open tube, $L = n\lambda/2$, i.e., $\lambda = 2L$, we can estimate the resonant frequency for different body features.

Eyeballs

The transverse diameter of the adult eyeball ranges from 2.1 to 2.7 cm [5]. Taking $\lambda = 2L = 2(0.024 \text{ m}) = 0.048 \text{ m}$:

$$f = \frac{v}{\lambda} = \frac{3.4 \times 10^7 \text{ m/s}}{2(0.024 \text{ m})} \approx 7.08 \times 10^8 \text{ Hz} \approx 708 \text{ MHz}.$$

Thus, for eyeballs, resonance occurs at much higher frequencies than the RF range of 80–100 MHz, which makes sense given the small eyeball diameter (about 33 times smaller than the resonant wavelength for 80–100 MHz).

Human Head

The average head circumference is approximately 55 cm for females and 57 cm for males [6]. Using an estimate of 56 cm, the diameter is:

$$d = \frac{56 \text{ cm}}{\pi} \approx 17.8 \text{ cm}.$$

$$f = \frac{3.4 \times 10^7 \text{ m/s}}{2(0.178 \text{ m})} \approx 9.5 \times 10^7 \text{ Hz} \approx 95 \text{ MHz}.$$

Since the head diameter is about a factor of 2 of the resonant wavelengths for 80–100 MHz, the corresponding frequency falls within this range, as confirmed above.

Infants

Infants are often between 47 and 53 cm in length [7]. Using $\lambda = 50 \text{ cm}$:

$$f = \frac{3.4 \times 10^7 \text{ m/s}}{2(0.50 \text{ m})} = 3.4 \times 10^7 \text{ Hz} \approx 34 \text{ MHz}.$$

Thus, the whole-body resonance frequency for infants is below the 80–100 MHz RF range by more than a factor of two. As babies grow, their whole-body lengths will correspond to frequencies farther from 80–100 MHz, though features of their bodies may then correspond more closely to the resonant wavelength [8]. For instance, the average height of a six-year-old

child is about 106 to 119 cm. Using $\lambda = 113$ cm:

$$f = \frac{3.34 \times 10^7 \text{ m/s}}{2(1.13 \text{ m})} \approx 1.5 \times 10^7 \text{ Hz} \approx 15 \text{ MHz}.$$

Considering these findings together, the radiation at these frequencies will be non-ionizing — meaning the photon energy $E = hf$ is millions of times too small to break chemical bonds. The only plausible effect is mild heating, which regulatory safety limits already restrict through the SAR standard of 1.6 W/kg in the USA.

Takeaways, Questions, and Concerns

Our biggest takeaway from this project is that there needs to be a lot more research conducted on the subject of cell phone and cell phone tower safety, especially since it has now become an integral part of modern human life. As a group we believe scientists should share their focus on both cell phone towers and cell phones themselves, as they both could pose a risk to the health and safety of all people. Because of the lack of research, people should not blindly believe that phones are completely safe or that phones are completely unsafe, but should make their own assessments on how they want to go about living their life and what cell phone safety means to them. This could mean only calling people on speaker phone or exclusively using wired headphones, but that is up to each person to decide what risks they want to take with their phones.

One question is how confidently we can talk about whole-body resonance when the human body consists of different types of tissue with no uniform geometry or material. The model of human tissue as like-water assumes that all tissue has the same water content, which is not true. The HPS article states that strong resonance occurs in the range of 80–100 MHz, which seems unlikely for all tissue to resonate together at one frequency. If this resonance value is accurate, it might represent an approximate range in which absorption is more effective rather than one defined peak. Furthermore, the 80–100 MHz RF absorption measurement corresponds to a standing human, so we are also curious about how RF energy absorption varies with changing positions such as sitting or lying down.

Overall, resonance can explain certain absorption peaks in the RF spectrum, but those peaks occur far below the microwave frequencies used for cellular communication. The main takeaway is that while the body can resonate with some radio frequencies, cell phone radiation is too high in frequency to cause whole-body resonance and too low in energy to cause ionization.

References

- [1] Classic, Kelly. *Radiofrequency (RF) Radiation*. Health Physics Society, 2025. <http://remm.hhs.gov/radiationspectrum.htm>
- [2] U.S. Department of Health and Human Services. *Electromagnetic Radiation: Ionizing and Non-ionizing*. Radiation Emergency Medical Management (REMM), updated 2 Oct. 2025. <https://remm.hhs.gov/radiationspectrum.htm>
- [3] American Cancer Society. *Do Cell Phones Cause Cancer? Cellphones and Cancer*, 2022. <https://www.cancer.org/cancer/risk-prevention/radiation-exposure/cellular-phones.html>
- [4] Federal Communications Commission. *Cell Phones and Specific Absorption Rate*. <https://www.fcc.gov/general/cell-phones-and-specific-absorption-rate>
- [5] Bekerman, Inessa, et al. Variations in eyeball diameters of the healthy adults. *Journal of Ophthalmology*, vol. 2014, p. 503645, 2014. <https://doi.org/10.1155/2014/503645>
- [6] *Human Head*. Wikipedia, 27 Sep. 2025. https://en.wikipedia.org/w/index.php?title=Human_head&oldid=1313653963
- [7] *What Is the Average Baby Length? Growth Chart by Month*. Medical News Today, 18 Mar. 2019. <https://www.medicalnewstoday.com/articles/324728>
- [8] *Normal Growth Rates for Kids*. Children's Wisconsin. <https://childrenswi.org/medical-care/adolescent-health-and-medicine/issues-and-concerns/adolescent-growth-and-development/normal-growth>
- [9] *Normal Liver Size by Age & Gender*. Asian Heart Insti-

tute, 25 Sep. 2024. <https://asianheartinstitute.org/blog/normal-liver-size-what-is-the-average-measurement-in-centimeters/>